

10.1 - Percentages

Problem Solving, Math 1000

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1 Definition

A **percentage** is a way of writing a fraction without having to deal with fractions. When we write " $x\%$ ", what we mean is " $\frac{x}{100}$ ". percentages are useful in several ways. Firstly, they serve as a common yardstick for measuring ratios and proportions of all types. Secondly, they provide a useful way of dealing with things like fees, taxes, and tips. And, finally, they help us better understand how things increase or decrease relative to some given baseline.

2 Observe

There are a few common "tips" we may wish to employ when working with percentages:

1. To express the ratio $\frac{A}{B}$ as a percentage, write the ratio as a decimal (that is, carry out the division of A by B) and multiply by 100.
2. To express $P\%$ as a decimal, divide P by 100. Alternatively, it is easy to do this simply by moving the decimal point two places to the left.
3. To find $P\%$ of B , divide P by 100 and multiply the result by B .
4. If you start with a baseline quantity B and want to add $P\%$ to it, you end up with a value (let's call it F) that is equal to $(100 + P)\%$ of B . That is, F (the result of adding $P\%$ to B) is $F = (1 + \frac{P}{100})B$.
5. If you have a value that you know is $P\%$ added to a baseline value B (let's call it F again) and you want to find B , you may use the previous formula to solve for B . This gives you $B = \frac{F}{1 + P/100}$.
6. Similarly to the last two instances, if you have a baseline value B and you know the value of F , we can find the percentage $P\%$ we added to B to get F by the following formula: $P\% = (100)\frac{F-B}{B}$

Notice that items 5 and 6 really just follow from some manipulations to the formula $F = (1 + \frac{P}{100})B$ in item 4, so you really only need to remember one equation.

3 Note

It is important to be careful for a few common errors people make when dealing with percentages.

Firstly, people often add percentages when it is not appropriate. For example, if you get a 10% raise one year and a 15% raise the next year, your total raise is more than $10\%+15\%=25\%$. This is because your second year's 15% raise is a 15% increase on your salary *after* the 10% raise of the first year has already been applied. There are a lot of other common things like this people might mistake. For example, if your stock portfolio goes up 25% one day and falls by 20% the next, you definitely didn't make any money. If you find shoes marked down 20% and they give you an additional 30% off the sale price, they aren't half price either.

The second mistake people often make with percentages is abusing very large numbers. While percentages above 100% are certainly a real thing, they often might not be employed in the correct way. For example, if an electronic store offers a 200% discount on a TV, you might ask the salesperson how they calculated it. The salesperson might say that the TV was originally \$ 3600 and is now on sale for \$ 1200, and since 3600 is a 200% increase from 1200, the TV must be on sale for 200% off. This is totally backwards, of course. The actual sale percent is $\frac{1200-3600}{3600} = 66.67\%$, and a 200% discount on a \$ 3600 TV would give it a net price of $-\$3600$ (that is, they would have to pay *you* the original price of the TV).